

Visual Inspect and Sticker Parts SOP

One episode inspects ten parts. It runs from ten parts standing in a numbered rack, a bare inspect plate under a lit lamp, three full dot strips, a blank grading card and three empty bins, to all ten parts looked at on all four faces, one dot sticker on every part, ten strokes on the card, six parts in the GOOD bin, two parts in the SCRATCH bin, two parts in the DENT bin and every tool back on its spot.

One part is finished before the next one is started. The order for each part never changes: **take the part out of its slot, stand it under the lamp, turn it and look four times, mark the card, put the dot on, drop the part in its bin.** No part is stickered until it has been looked at on all four faces, and no part goes in a bin until its stroke is on the card.

One arm does all of it. The same gripper takes every part out of the rack, stands it on the inspect plate, turns it, works the marker, presses the dot on and drops the part in its bin. Nothing in this task has to be held, because nothing moves on its own: the rack, the inspect plate, the sticker board, all three bins and the card all grip the table, the lamp stands on its own, and a part standing on the plate stands there on its own. Only one arm works in this cell. Do not bring a second arm into the task.

Every spot sits inside the arm's comfortable reach, so the arm never has to stretch or fold in on itself. Only one part is held at a time, no grasp is changed while a part is off the table, and no part is ever carried over a bin it is not going into or over the sticker board on its way somewhere else.

The lamp does the showing and the eye does the deciding. A part is looked at with the gripper open and away from it, so nothing is hidden behind a finger, and it is looked at on all four faces every time, even after a defect has already been found. Nothing is measured, no number is read, and no part is compared against another part.

The card is written in the plainest words the job allows. A part is not marked non conforming or cosmetically rejected: it is marked **SCRATCH** or **DENT**, and a part with neither is marked **GOOD**. Those same three words are printed on the three bins, so the word on the card, the colour of the dot and the bin the part lands in all say the same thing. Anyone who can read three words can read the card.

Each tool goes back to its own spot before the next step starts. The counts are fixed: ten parts, four looks per part, one dot per part, ten strokes on the card, six parts in the GOOD bin, two in the SCRATCH bin and two in the DENT bin. There is no second look after a part has been stickered and no second dot on the same part. A part that lands off the table is left exactly where it falls, tagged, and the episode continues. Nothing is ever picked up off the floor on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the parts rack, the inspect plate under its lamp, the sticker board with all three strips, all three bins, the grading card and the marker holder.
3. The four side faces of a part standing on the inspect plate stay in frame and lit for the whole of Step 3, so a reviewer can see the same face the arm was shown.
4. All ten rows of the grading card and all three of its columns stay in frame and readable the whole time.
5. The lamp is on and stays on for the whole episode. It lights the inspect plate evenly, throws no shadow of the arm across the part at any of the four stops, and throws no glare back into the camera.
6. The arm is at home with its gripper open.
7. The tabletop is dry and clear of anything but the items in the Materials checklist.
8. The arm reaches every spot on the table, all ten rack slots, the inspect plate, all three sticker strips, all three bins, the grading card and the marker holder, without extending to a joint limit or folding in on itself. Bring the spots forward or closer together until this is true.
9. The arm reaches every one of the ten rack slots straight on, so a part can be lifted straight up out of any slot without leaning it over.
10. The arm comes down onto the inspect plate straight from above and can close on a standing part from four sides, so the part can be turned at every stop without the arm reaching round the lamp stand.
11. Nothing the arm carries passes under the lamp head or through the lamp stand. The lamp is never touched, moved, switched or leaned on during the episode.
12. The paths from each rack slot to the inspect plate, from the inspect plate to each sticker strip, from the marker holder to the card, and from each sticker strip to each bin are unobstructed. Carry one part along each path by hand during setup: it must clear the lamp stand, the card and the rim of every bin it passes.

Materials checklist

1. One parts rack stands at the back of the table, straddling the midline, with ten upright square slots in two rows of five, numbered 1 to 5 across the back row and 6 to 10 across the front row, left to right. The numbers face the camera.
2. The rack grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or replace it. Nothing pins the rack during the episode.
3. Each slot holds one part standing upright with enough clearance that the part lifts straight out without rubbing, and close enough that the part stands square rather than leaning.
4. Ten parts, and no more. Each part is a plain square block, flat on the top and the bottom, with four flat side faces and no markings, no numbers and no dots on it at the start of the episode. Every part is the same size and the same colour, so the shape tells a reviewer nothing and only the lamp does.
5. Each part is tall enough for the gripper to close on two opposite side faces squarely, and wide enough on the bottom to stand on its own on the inspect plate without rocking.
6. The bottom face of every part is clean, dry and bare, wide enough to carry one dot sticker well inside its edges.
7. The ten parts are the same ten in every episode: six good parts, two parts with a scratch and two parts with a dent.
8. The four bad parts are bad by a plain amount. A scratch is one bright line running across one side face, long enough and bright enough to see across the table under the lamp. A dent is one dip in one side face, deep enough to hold its own shadow under the lamp. A part with a mark that only shows at one angle is not used.
9. Each bad part carries one defect only, on one side face only. No part carries both a scratch and a dent, and no good part carries any mark at all.
10. One inspect plate sits at the middle of the table, straddling the midline, directly under the lamp. It is flat, matt, and a plainly different colour from the parts. It is heavy and it grips the table.
11. The inspect plate carries one square outline in its middle, the size of a part's bottom face, so every part stands in the same place under the lamp in every episode. The plate is bare at the start of the episode and bare again at the end.
12. One lamp stands behind the inspect plate on its own stand, its head over the plate and its stand clear of the plate, of the rack and of every carry path. It is on before the episode starts and it stays on until the recording stops.
13. One sticker board lies to the front of the inspect plate, straddling the midline, holding three strips of dot stickers in three labelled channels: **GREEN** at the back, **YELLOW** in the middle and **RED** at the front. The labels face the camera.

14. Each strip holds ten dot stickers lying sticky side up in a straight row, each dot in its own marked place, the places numbered 1 to 10 left to right. Dots are taken left to right, so the next free dot is always the leftmost place that still has a dot on it.
15. The dots stick to a part's bottom face on one press and come away from the strip cleanly. Press one dot onto a spare part by hand during setup: if the dot tears, stays on the strip or lifts at an edge, replace the strip.
16. The sticker board grips the table and none of the three strips lifts, curls or slides under a press. Nothing pins the board during the episode.
17. Three bins stand in their holders, each one labelled on the side facing the camera: **SCRATCH** on the left of the sticker board, **DENT** on the right of it, and **GOOD** between them at the front. Each bin is empty and dry at the start of the episode, and the same bin carries the same word in every episode.
18. Each bin is deep enough that a part dropped in from just above the rim stays in it, and wide enough that a part goes in straight down without the gripper touching the rim.
19. One grading card lies at the front of the table, straddling the midline. It is printed with ten rows numbered 1 to 10 down its left side, one row for each rack slot, and three columns headed **GOOD**, **SCRATCH** and **DENT** across the top. Every box is empty at the start of the episode.
20. The card grips the table, and every box is large enough to hold one stroke well inside its lines without touching the next box.
21. One marker stands upright in its holder on the card's right edge, cap off, writing cleanly, and short enough for the gripper to close on it and still reach the bottom row of the card.

Workspace layout

Three rows: the parts at the back, the inspect plate under its lamp in the middle, and the sticker board with the three bins and the card at the front. Every part travels forward out of the rack to the inspect plate, forward again to the sticker board, then into a bin.

Back row:

1. Parts rack, back centre on the midline, ten upright square slots in two rows of five, slots 1 to 5 across the back row and slots 6 to 10 across the front row, left to right, with the numbers facing the camera.

Middle row:

2. Inspect plate, centre, straddling the midline, with the square outline in its middle and the lamp standing behind it, the lamp head over the plate and the lamp stand clear of every carry path. The plate is bare at the start of the episode and bare again at the end.

Front row, left to right:

3. SCRATCH bin spot, left of the sticker board, the bin standing in its holder with its label facing the camera.
4. Sticker board, front centre on the midline, carrying the GREEN strip at its back, the YELLOW strip in the middle and the RED strip at its front, all three labelled and facing the camera.
5. DENT bin spot, right of the sticker board, the bin standing in its holder with its label facing the camera.
6. GOOD bin spot, front centre, in front of the sticker board and between the other two bins.
7. Grading card, front right, its ten numbered rows and its three column headings facing the camera.
8. Marker holder, on the grading card's right edge.

A clear gap of at least a gripper's width sits between the inspect plate and the lamp stand, between the sticker board and each bin, and between the three strips on the board, so a part comes down onto one strip without touching another, and drops into a bin straight from above without the gripper touching the board or the bin rim. No carry passes over a bin the part is not going into, over the lamp head, or over a strip the part is not going onto.

Vocabulary

- **Arm assignment:** one gripper does everything: every part, every turn, every dot, every mark. This is a single arm task and no second arm takes any part in it.
- **Tool home:** the spot a tool returns to and sits in before the next step starts: the marker holder. A tool is never left on the inspect plate, on the card, on the sticker board, in a bin or in the gripper between steps.
- **Release point:** the gripper opens over the square outline on the inspect plate when a part is being stood under the lamp, over the sticker board when a dot is being pressed on, just above the rim of a bin when a part is being dropped in, or over a tool home. Never over the wrong bin, never over the lamp and never over open tabletop.
- **Stable placement:** the object stands still after release, square on its surface, and does not rock, tip, slide or lean on anything.
- **Part:** one plain square block with four flat side faces. There are ten in every episode, one to a rack slot.
- **Side face:** one of the four flat faces around a part. All four are looked at on every part, and a defect is only ever on one of them.
- **Inspect plate:** the flat plate under the lamp that a part stands on while it is looked at. Only one part stands on it at a time.
- **Square outline:** the printed square in the middle of the inspect plate. A part stands inside it, so every part is lit the same way.

- **Lamp:** the fixed light over the inspect plate. It is on for the whole episode and it is never touched, moved or switched by the arm.
- **Quarter turn:** one quarter of a full turn about the part's own upright axis, which brings the next side face round to the camera. The part stays standing the whole way, and it is never tipped on its side or turned over in the air.
- **Stop:** one of the four positions a part stands in on the plate, one for each side face. Every part is looked at at all four stops and comes back to the face it started on.
- **Look:** the gripper open, taken right away from the part, and the part standing still under the lamp on its own for 2 seconds with nothing over it. A face that was seen with the gripper still on the part has not been looked at.
- **Clean face:** a side face with no bright line and no dip on it under the lamp, at that face's own stop.
- **Scratch:** one bright line running across a side face. The plain word for it is **SCRATCH**.
- **Dent:** one dip in a side face that holds its own shadow under the lamp. The plain word for it is **DENT**.
- **Good part:** a part whose four side faces were all clean. It is marked **GOOD**, it takes a green dot and it goes in the GOOD bin.
- **Defect code:** the one plain word written on the card for a part: **GOOD**, **SCRATCH** or **DENT**. There are three codes and no others, and no part gets two of them.
- **Dot sticker:** one coloured dot from a strip, pressed onto the bottom face of a part. Green says GOOD, yellow says SCRATCH, red says DENT. One dot per part.
- **Dot strip:** one row of ten dot stickers lying sticky side up in its own labelled channel on the sticker board. Dots are taken from the left end first.
- **Next free dot:** the leftmost place on a strip that still has a dot on it. It is the only dot a part is ever pressed onto.
- **Bare place:** a place on a strip whose dot has gone onto a part. It is how a reviewer counts the dots that have been used.
- **Marked box:** a box on the card holding one stroke, fully inside its lines and not touching the next box.
- **Tally stroke:** one straight upright stroke inside one box. One part draws one stroke, in one row and one column.
- **Graded bin:** one of the three bins, each carrying one of the three codes on its side: GOOD, SCRATCH or DENT. A part goes in the bin whose word matches the word on its card row.
- **One part at a time:** the gripper takes one part out of the rack, looks at it, marks it, stickers it and bins it before it touches the next part.
- **Bare plate:** nothing on the inspect plate and nothing inside the square outline.

- **Spill:** a part outside the rack, the inspect plate, the sticker board and all three bins. It is left where it falls and cleared in the reset, never on camera.

Steps

Step 1: Check the lamp, the strips and the card

Goal: the lamp lit and steady over a bare inspect plate, all three strips full, and every box on the card empty.

This check happens once, before the first part. If the lamp is dim or throws a shadow across the plate, every look in the episode is worth less, so the light is proved before anything is inspected.

1.1 Check the light on the plate

- **Check:** the inspect plate is bare, with nothing inside the square outline.
- Bring the **gripper** to the height it will work at over the plate, hold still for 2 seconds, then take it right away.
- The plate is evenly lit with the arm clear of it, and the arm throws no shadow across the square outline when it is at working height.

Check: the square outline is lit evenly all the way round and no shadow of the arm falls inside it. If the lamp is dim, flickering or shadowing the outline, end the episode and correct the lamp during reset. The arm never touches the lamp to fix it.

1.2 Check the three strips

- **Check:** all three strips are full, ten dots in ten places on each, with no bare place anywhere and no dot lifted at an edge.
- **Check:** the GREEN, YELLOW and RED labels are readable in frame.

Check: every strip is full and flat. If a strip is short of ten dots, or a dot is curled, torn or missing, end the episode and replace the strip during reset.

1.3 Check the card and the bins

- **Check:** every box on the card is empty, in all ten rows and all three columns.
- **Check:** the GOOD, SCRATCH and DENT bins each stand upright in their holders, empty and dry, with their words facing the camera.
- **Check:** the marker stands upright in its holder with its cap off.

Check: the card starts blank, the three bins start empty, and each bin's word matches a column heading on the card. If a box is already marked or a bin is not empty, end the episode and reset instead. One check of the lamp, the strips and the card, and one only. It is not run again between parts.

Step 2: Take the part from its slot

Goal: one part out of its slot, hanging level in the gripper, and its slot empty.

Parts are worked in slot order, slot 1 first and slot 10 last. The rack and the inspect plate both grip the table, so nothing holds either of them while the part is moved.

2.1 Lift the part straight out

- With the **gripper**, close on two opposite side faces of the part square, and lift it straight up out of the slot until it is clear of the rack.
- Keep it upright and level. Do not lean it over on the way out.
- The gripper closes on the side faces only. It never covers the bottom face, because the dot goes there.

Check: the part hangs level in the gripper and does not slip. If it hangs tilted, stand it back in its slot and take it again. Make no more than two corrective takes. If it still will not hang level, end the episode and correct the part during reset.

2.2 Stand the part under the lamp

- Carry the part forward to the inspect plate, keeping it upright and clear of the lamp stand, and bring it over the square outline.
- Lower it straight down inside the outline until it takes its own weight, open the gripper, and take the gripper right away from the part and the plate.

Check: the part stands square inside the outline, upright, touching nothing but the plate, and does not rock. If it stands outside the outline or rocks, lift it clear and stand it again. Make no more than one correction.

Step 3: Turn the part and look, four times

Goal: all four side faces of the part seen under the lamp, each one with the gripper open and away, and the part back on the face it started on.

The part is let go of at every stop. It is never looked at while it is still in the gripper, because a gripper on a part covers part of two faces and can hide the very mark the look is for.

3.1 Look at the face in front

- The gripper is open and right away from the part, and nothing is over the plate.
- Hold still and wait 2 seconds with nothing touching the part.
- **If the face carries a bright line running across it, the part is SCRATCH.**
- **If the face carries a dip that holds its own shadow, the part is DENT.**
- **If the face carries neither, the face is clean and the look moves on.**

The camera records what was on the face. It is not written down anywhere but the card, and it is written down in Step 4.

3.2 Turn a quarter turn

- With the **gripper**, close on two opposite side faces of the part square, still inside the square outline.
- Turn the part a quarter turn about its own upright axis, keeping it standing on the plate the whole way, until the next side face comes round to the camera.
- Lower it back square inside the outline, open the gripper, and take it right away.
- The part is never lifted off the plate for a turn and never tipped onto its side.

Check: the part stands square inside the outline again, one quarter turn on from where it was, and has not been dragged outside the outline or knocked over.

3.3 Repeat to four stops

- Repeat 3.1 and 3.2 until all four side faces have been looked at.
- Four stops in all, four quarter turns in all, so the part finishes on the face it started on.
- Nothing else changes between the stops. The lamp is not touched and the plate is not moved.

Check: four looks have been taken, each with the gripper open and away for 2 seconds, and the part is back on its starting face. A part is not looked at a fifth time and a face is not looked at twice. If a face was missed, end the episode and let review flag it.

3.4 Read the part

- If all four faces were clean, the part is **GOOD**.
- If one face carried a bright line, the part is **SCRATCH**.
- If one face carried a dip, the part is **DENT**.

One code per part, and one only. A part with a mark on it is not weighed up against another part and is not called a near miss: it takes the plain word for what was on it.

Check: the part has one code and the part is still standing inside the outline, untouched, ready for Step 4.

Step 4: Mark the card

Goal: one stroke on the card, in the row for this part's slot number and in the column for its code, with the marker back in its holder.

The stroke goes on the card while the part is still standing under the lamp, so a reviewer can see the face and the mark in one frame. The part is not touched again until the card is marked.

4.1 Take the marker

- With the **gripper**, close on the marker, lift it straight up out of its holder, and carry it to the card.
- Keep it upright, tip down, the whole way.

Check: the marker hangs upright in the gripper and the inspect plate is untouched.

4.2 Draw one stroke

- Bring the tip over the row for this part's slot number and over the column for its code: **GOOD**, **SCRATCH** or **DENT**.
- Draw one straight upright stroke inside that box, well inside its lines and not touching the next box.
- Lift the tip clear of the card.
- One stroke for this part, in one box. No second stroke and no crossing out.

Check: the stroke sits inside its own box, in the right row and the right column, and no other box has been marked. If the stroke has run outside its box or into the next one, leave it and let review flag the episode. A stroke is never wiped off or drawn over on camera.

4.3 Park the marker

- With the **gripper**, carry the marker back to its holder, lower it until it stands on its own, release, and lift clear.

Check: the marker stands upright in its holder, the card carries one more stroke than it did, and the part is still standing inside the outline.

Step 5: Put the dot on

Goal: one dot sticker on the bottom face of the part, taken from the strip that matches its code, and one more bare place on that strip.

The colour of the dot says the same thing the card says. A green dot goes on a GOOD part, a yellow dot on a SCRATCH part and a red dot on a DENT part.

5.1 Lift the part off the plate

- With the **gripper**, close on two opposite side faces of the part square, and lift it straight up off the inspect plate until it is clear of the plate.
- Keep it upright and level, bottom face down.

Check: the inspect plate is bare and the part hangs level in the gripper.

5.2 Press the part onto the next free dot

- Carry the part forward to the sticker board and bring it over the strip that matches its code: **GREEN** for GOOD, **YELLOW** for SCRATCH, **RED** for DENT.
- Line the part up over the next free dot, the leftmost place on that strip that still has a dot on it.
- Lower the part straight down onto the dot until its bottom face is flat on the strip, hold for 2 seconds, then lift it straight up again.
- Do not slide, twist or rock the part on the strip. Straight down, hold, straight up.

- The part never passes over a strip it is not going onto.

Check: the place on the strip is bare and the dot has come away with the part. If the dot is still on the strip, lower the part onto the same place and press it once more. Make no more than one correction. If the dot still will not come away, end the episode and replace the strip during reset.

5.3 Check the dot

- Hold the part still above the board for 2 seconds so the strip below it is readable in frame.

Check: one dot has gone from the strip, its colour matches the column marked on the card for this part, and no dot has come off any other place on any strip. One dot per part, and one only. A part is never pressed onto a second dot.

Step 6: Drop the part in its bin

Goal: the part in the bin whose word matches its code, and the gripper clear and empty.

The word on the bin, the word on the card and the colour of the dot all say the same thing. A part goes straight from the sticker board into its bin, without passing over either of the other two.

6.1 Carry the part to its bin

- A **GOOD** part goes to the **GOOD** bin, front centre.
- A **SCRATCH** part goes to the **SCRATCH** bin, on the left of the sticker board.
- A **DENT** part goes to the **DENT** bin, on the right of the sticker board.
- Carry the part level, clear of the sticker board and clear of the rims of the other two bins, and bring it over the middle of its own bin.

Check: the part is over the bin whose word matches the column marked on the card, and has passed over no other bin on the way.

6.2 Let it go

- Lower the part until it is just above the bin rim, open the gripper, and let it drop in.
- Lift the gripper clear of the bin and away from the board.
- The part is not thrown, and the gripper does not go down inside the bin.

Check: the part is in its bin, the gripper is empty and clear, and the bin still stands upright in its holder. If the part has landed outside a bin, leave it where it fell and let review flag the episode. Nothing is picked up off the floor on camera.

Step 7: Work through all ten parts

Goal: all ten slots empty, ten strokes on the card, ten bare places across the three strips, and ten parts in the three bins.

- Repeat Steps 2 to 6 for every part, slot 1 first and slot 10 last.

- One part at a time. The next part is not taken out of the rack until the last one is in its bin.
- The inspect plate is bare between parts, and the marker is in its holder between parts.
- The counts come out at six parts in the GOOD bin, two in the SCRATCH bin and two in the DENT bin, with six green dots, two yellow dots and two red dots gone from the strips.

A part is never gone back to. Once a part is in a bin it is not taken out again, not looked at again and not stickered again, whatever a later part shows.

Check: after each part, the rack has one fewer part in it, the inspect plate is bare, one more stroke is on the card, one more place is bare on a strip, and no part is inspected twice or skipped.

Step 8: End the episode

- Confirm the rack: all ten slots empty.
- Confirm the inspect plate: bare, with nothing inside the square outline.
- Confirm the card: ten strokes in all, one in every row from 1 to 10, six in the GOOD column, two in the SCRATCH column and two in the DENT column, each stroke inside its box and standing clear of the next.
- Confirm the strips: six bare places at the left end of the GREEN strip, two at the left end of the YELLOW strip and two at the left end of the RED strip, and no bare place anywhere else.
- Confirm the bins: six parts in the GOOD bin, two parts in the SCRATCH bin, two parts in the DENT bin, all three upright in their holders.
- Confirm the tools: the marker upright in its holder, the lamp still on and untouched.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return the arm home with its gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the parts out of all three bins and stand the bins back in their holders, empty and dry, with their words facing the camera.
2. Peel the dot off the bottom face of every part and bin the dots. Wipe any glue off the bottom face and dry it, so every part starts the next episode bare.
3. Count the parts back into the rack: exactly ten, six good, two scratched and two dented. Replace any part that has picked up a new mark, so the ten are the same ten next time.
4. Shuffle the ten parts into the ten slots so they do not sit in the same order twice in a row, then stand each one upright and square, and turn each one so the marked face does not start toward the camera in every episode.

5. Pick up any part that fell off the table or landed outside a bin and put it back in the rack. Replace it if it is damaged.
6. Wipe the grading card clean and dry it, so all ten rows and all three columns start blank.
7. Lay three fresh strips in the sticker board channels, ten dots to a strip, GREEN at the back, YELLOW in the middle and RED at the front, every dot sticky side up and flat in its own place.
8. Press one spare dot onto a spare part by hand and confirm it comes away from the strip cleanly. Replace a strip whose dots tear, curl or stay put.
9. Wipe the inspect plate clean and dry, and confirm the square outline is still readable.
10. Confirm the lamp is on, steady and clean, its head still over the plate and its stand still clear of every carry path. Replace a bulb that is dim or flickering.
11. Stand the marker in its holder with the cap off. Replace a marker that is running dry.
12. Confirm the rack, the inspect plate, the sticker board, all three bins and the card all still grip the table.
13. Confirm the card is still printed in full plain words, with GOOD, SCRATCH and DENT readable and the ten rows still numbered 1 to 10.
14. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a part is stickered before all four faces have been looked at, a part goes in a bin before its stroke is on the card, the dot goes on before the card is marked, a new part is taken while the last one is still on the plate or still in the gripper, the lamp and the card are not checked before the first part, or a step is skipped.

- **SOP rule broken:** Steps 1 to 7, check the lamp, the strips and the card once, then for each part take it, stand it under the lamp, look at four faces, mark the card, sticker it, bin it, in that order.
- **Coaching note:** light first, then one part all the way through. The stroke goes on the card while the part is still under the lamp.

Violation: A second arm used

- **Visible cue:** a second arm enters the work area, closes its gripper, touches an object, steadies the rack, the inspect plate, the sticker board, a bin or the card, or moves in front of the lamp, the plate or the card. Any object taken or carried by a second gripper is a violation on its own.
- **SOP rule broken:** Steps 1 to 8, this is a single arm task: one gripper does all of it and nothing else in the cell touches the task.
- **Coaching note:** one arm only. If a second arm has worked, the episode is a bimanual episode wearing the wrong label.

Violation: Object held or steadied

- **Visible cue:** the gripper presses on the rack, the inspect plate, the sticker board, a bin, the card or the lamp stand to keep it still while working, or holds the part on the plate while a face is being looked at, instead of letting the plate do it.
- **SOP rule broken:** Steps 1 to 7, nothing is held in this task: every fixture grips the table and a part standing on the plate stands on its own.
- **Coaching note:** let it go and come away. A hand on the part is a hand over the face being looked at.

Violation: Part looked at in the gripper

- **Visible cue:** a face is shown to the camera with the gripper still closed on the part, the gripper stays over the part during the 2 seconds, the part is lifted off the plate to be looked at, or the wait is cut short before 2 seconds are up.
- **SOP rule broken:** Step 3.1, the gripper is open and right away from the part, and the part stands still on its own for 2 seconds at every stop.
- **Coaching note:** open, away, wait. A face seen past a gripper is a face that was not looked at.

Violation: Face count wrong

- **Visible cue:** fewer or more than four looks are taken on a part, a face is looked at twice, a face is never brought round to the camera, or the part does not finish on the face it started on.
- **SOP rule broken:** Step 3.3, four stops and four quarter turns per part, all four side faces, back to the starting face.

- **Coaching note:** the count is fixed at four. Every part goes all the way round, even one that showed its mark on the first face.

Violation: Turn done wrong

- **Visible cue:** the part is lifted off the plate to turn it, tipped onto a side face, turned over, turned more or less than a quarter turn between stops, dragged outside the square outline, or spun round so fast that no face stops square to the camera.
- **SOP rule broken:** Step 3.2, the part is turned a quarter turn about its own upright axis, standing on the plate the whole way, and stood back square inside the outline.
- **Coaching note:** turn it on its feet, one quarter at a time. Nothing on this station is picked up to be turned.

Violation: Wrong code marked

- **Visible cue:** the stroke goes in a column that does not match what the lamp showed, in a row that is not this part's slot number, or a code other than GOOD, SCRATCH or DENT is written on the card.
- **SOP rule broken:** Step 4.2, one stroke in the row for this part's slot number and in the column for the code the four looks gave.
- **Coaching note:** three words, and only three. Mark the row the part came out of, not the row after it.

Violation: Card marked wrong or marked twice

- **Visible cue:** more than one stroke is drawn for a part, a stroke runs outside its box or into the next one, a stroke is crossed out, wiped off or drawn over, or the marker touches the card before the four looks are finished.
- **SOP rule broken:** Step 4.2, one stroke per part, inside its own box, and no correction on camera.
- **Coaching note:** one stroke, one box. A card that has been corrected on camera cannot be read back.

Violation: Dot colour wrong

- **Visible cue:** the part is pressed onto a strip whose colour does not match the column marked on the card: a green dot on a SCRATCH or DENT part, or a yellow or red dot on a GOOD part.
- **SOP rule broken:** Step 5.2, green for GOOD, yellow for SCRATCH, red for DENT, matching the column marked on the card.
- **Coaching note:** the dot and the card say the same thing. Read the column just marked, then go to that strip.

Violation: Dot count or dot place wrong

- **Visible cue:** a part is pressed onto more than one dot, two parts are pressed onto the same place, a dot is taken from somewhere other than the leftmost place still holding a dot, a dot is left half on the strip and half on the part, or a part goes to its bin with no dot on it.
- **SOP rule broken:** Steps 5.2 and 5.3, one dot per part, taken from the next free dot at the left end of the strip, pressed on whole and lifted away whole.
- **Coaching note:** work the strip from the left end. One press, one dot, and check it has come away before the carry.

Violation: Sticker pressed wrong

- **Visible cue:** the part is slid, twisted or rocked on the strip, pressed on somewhere other than its bottom face, pressed on with the gripper covering the dot's place, or lifted before the 2 second press is up.
- **SOP rule broken:** Step 5.2, straight down onto the dot, flat on the bottom face, hold 2 seconds, straight up.
- **Coaching note:** down, hold, up. A dot dragged sideways lifts at an edge and comes off in the bin.

Violation: Part in the wrong bin

- **Visible cue:** a part goes in a bin whose word does not match the column marked on the card, or whose word does not match the colour of the dot on the part.
- **SOP rule broken:** Step 6.1, the word on the bin, the word on the card and the colour of the dot all match.
- **Coaching note:** three things say the same thing before the gripper opens. If two of them disagree, the part is in the wrong place.

Violation: Part dropped from too high, or thrown

- **Visible cue:** the gripper opens well above the bin rim so the part falls a visible distance, the part is flicked or thrown into the bin, the part bounces out of the bin, or the gripper goes down inside the bin to place it.
- **SOP rule broken:** Step 6.2, lower the part until it is just above the bin rim, then open the gripper.
- **Coaching note:** bring it to the rim and let it go. A part that bounces out was let go too high.

Violation: Lamp touched or moved

- **Visible cue:** the arm touches, knocks, leans on, switches or shades the lamp or its stand, or the lamp head is moved so the square outline is lit differently from one part to the next.
- **SOP rule broken:** Steps 1 to 7, the lamp is on before the episode and is never touched by the arm.

- **Coaching note:** the light stays put. Every part is lit the same way or the looks cannot be compared.

Violation: Bad pick, carry, or release

- **Visible cue:** a gripper shifts or seats its hold again while a part is off the table, a part or the marker slides along the table with no daylight under it, a part hangs tilted in the gripper because it was taken off square, the gripper closes over the bottom face of a part, or a part is tipped over or turned over in the air.
- **SOP rule broken:** Steps 2 to 6, one part at a time, closed on two opposite side faces square, lifted straight up clear of the surface before any horizontal motion, and carried upright.
- **Coaching note:** set it down to change the grip, and lift before the carry. A part that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** a part, a bin, the card, the sticker board or the inspect plate is shoved, nudged or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 7, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Tool not returned to its home

- **Visible cue:** the marker is left on the card, on the inspect plate, on the sticker board or in a bin, the gripper opens while the marker is still above its holder, or the marker is still in the gripper when the next step begins.
- **SOP rule broken:** Step 4.3, the marker returns to its holder and stands upright in it before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower it until it stands, then release.

Violation: Part gone back to

- **Visible cue:** a part is taken back out of a bin, looked at again, turned again, marked again or stickered again after it has been binned, or a stroke is added to the card for a part that is already in a bin.
- **SOP rule broken:** Step 7, once a part is in a bin it is finished, whatever a later part shows.
- **Coaching note:** the episode moves forward only. A part revisited makes the record unreadable.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, the gripper or a carried part strikes the rack, the lamp stand, the inspect plate, the sticker board, a bin or the card and knocks, tips, slides or displaces it; a part is knocked out of another slot on the way out of the rack; or a part falls short of its bin, with no gripper failure, drift, or controller fault visible.
- **SOP rule broken:** Steps 1 to 7, keep each part on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. The lamp stand and the bin rims are the two things most often clipped.

Violation: Manipulation outside the camera frame

- **Visible cue:** a pick, a turn, a look, a stroke, a press or a drop happens partly or wholly outside the environment camera frame, or the arm stands between the camera and the part, the strip or the card while a face is being looked at, a dot is being pressed on or a stroke is being drawn.
- **SOP rule broken:** Steps 1 to 7, every manipulation happens inside the camera frame and the part, the strips and the card stay visible.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions and keep the arm clear of the plate and the card.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a part still in the rack or still on the inspect plate, a stroke missing or an extra stroke on the card, a part with no dot on it, a part in the wrong bin, the marker out of its holder, the arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 8, confirm the rack, the plate, the card, the strips, the bins and the tools, then return the arm home with its gripper open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arm does before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside the arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.

- Lamp fault: a bulb that goes out, dims or flickers during the episode, or a lamp head that droops or turns on its own so the plate stops being evenly lit.
- Defective inspect plate: one that slides, turns or lifts on the table under a correct placement, that is not flat, or whose square outline has worn away.
- Defective rack: one that slides, turns or lifts under a correct pick, or a slot that grips a part so a straight lift will not free it.
- Defective sticker strip: one whose dots tear, curl at an edge, stay on the strip under a correct press, or come away two at a time.
- Sticker board that lifts, slides or curls under a correct press.
- Defective bin: one that will not stand upright in its holder, or that is dented or cracked so a part bounces back out of it.
- Dry marker, or a grading card that will not wipe clean so a box or a row starts the episode already marked.
- A grading card printed wrong: a missing row number, a missing column heading, columns headed other than GOOD, SCRATCH and DENT, rows numbered other than 1 to 10, or boxes too small to hold a stroke clear of the next box.
- A part whose mark is so faint that it neither plainly shows nor plainly does not, so no one can call it. Replace the part before the next episode.
- A new scratch or dent picked up during handling that was not on the part at the start of the episode.
- Glue, dust or grit left on a part's bottom face from a previous episode, so the dot will not stick.
- Rack holding fewer or more than ten parts at episode start, or the wrong mix of them. Reset error, not a violation.
- A strip holding fewer than ten dots at episode start. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take one part from its rack slot
2. Stand one part on the inspect plate
3. Look at one face under the lamp
4. Turn the part a quarter turn
5. Take the marker from its holder
6. Draw one stroke on the grading card
7. Return the marker to its holder
8. Lift one part off the inspect plate
9. Press one part onto a dot sticker

10. Drop one part in the GOOD bin
11. Drop one part in the SCRATCH bin
12. Drop one part in the DENT bin
13. Return the arm home and end the episode